



Introduction to Macroeconomics

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Problem Set 4

2025

1 Inflation

Exercise 1

Assuming that prices are flexible, the Quantitative Theory of Money states that variations in money supply only affect nominal variables. In this case, we say that there is 'money neutrality' on real variables in the long run. Let's suppose that in a given economy the money demand is given by :

$$\left(\frac{M}{P}\right)^d = cY$$

1. If the quantity of output is $Y = 300$ and the velocity of money is $V = 2$, what is the value of ' c '? Can you give an interpretation for the parameter ' c '?
2. Use the Quantitative Theory of Money to derive the relationship between the growth rate of money, economic growth and the inflation rate.
3. If V remains constant over time and the GDP grows at an annual rate of 2%, is there a growth rate of money that would allow the inflation rate to be zero?

Exercise 2

Countries that use inflation as a tax decide to do so because :

- (a) The government does not understand the causes and consequences of inflation.
- (b) Public spending is high, the government does not have enough revenues and government borrowing is limited.
- (c) An inflation tax does not create economic distortions.
- (d) An inflation tax is the most equitable tax that you could impose on economic agents.

2 Open Economy

Exercise 3

We have a world where there are only two economies : Country A and Country B :

Country A		Country B	
Consumption	200	GDP	300
Investment	50	Consumption	180
Taxes	50	Net exports	30
		Budget deficit	0

We know that in Country A the price level is very high, and therefore its citizens use all their income to consume, so private saving is zero. We also know that 20% of the world output is saved. Which of the following propositions is correct ?

- (a) The total investment in the world is equal to 110, the public spending of Country A and Country B are the same and the investment in Country B is equal to 90.
- (b) The GDP of country A is equal to its consumption because the savings in Country A are equal to 0.
- (c) The total investment in the world is equal to 110, the public spending of Country A and Country B are the same and the investment in Country B is equal to 60.
- (d) The total investment in the world is equal to 105, the budget deficit of Country A is higher than that of Country B and the investment in Country B is equal to 90.

Exercise 4 (Exam June 2013)

Take the following statistics :

	Price of a Big-Mac in local currency	Nominal exchange rate on the currency market
USA	USD 3	-
Japan	Yen 270	121 Yen = 1 USD
China	Yuan 12	7.8 Yuan = 1 USD
Switzerland	CHF 6	1.25 CHF = 1 USD

According to the Big Mac index, which of the following is the case ?

- (a) One can expect an appreciation of all three currencies with respect to the dollar.
- (b) The Swiss franc is overvalued with respect to the dollar.
- (c) The Yen and the Yuan are both overvalued with respect to the dollar.
- (d) One can expect an appreciation of the dollar with respect to the Yuan.

Exercise 5 (Exam June 2014)

According to the relative PPP theory for the determination of exchange rates, if price levels in China increase at an annual rate of 8% , and at an annual rate of 2% in Switzerland, we can conclude that there will be :

- (a) a 6% appreciation of the CHF with respect to the Yuan.
- (b) a 6% depreciation of the CHF with respect to the Yuan.
- (c) a 10% appreciation of the CHF with respect to the Yuan.
- (d) a 10% depreciation of the CHF with respect to the Yuan.

Exercise 6

Let's assume that in France it becomes fashionable to consume Swiss cheese. Please answer to the following questions using a graphical approach. Consider Switzerland as the domestic country .

1. What is the effect of this phenomenon on the demand for Swiss Francs in the market for (domestic) currency ?
2. What happens to the value of the Swiss Franc in the market for (domestic) currency ?
3. What happens to the net exports and to the net capital outflow ?

3 Additional questions - Not treated in seminar

Exercise 7 (Exam August 2014)

According to open macroeconomics model seen in class, if a government returns to a balanced budget after being in deficit, you can expect that :

- (a) The supply of loanable funds increases, decreasing the interest rate and increasing the net capital outflow, which implies a depreciation of the national currency and increase in net exports.
- (b) The demand for loanable funds increases, increasing the interest rate and decreasing the net capital outflow, which implies an appreciation of the national currency and an increase in net exports.
- (c) The supply of loanable funds increases, decreasing the interest rate and increasing the net capital outflow, which implies an appreciation of the national currency and a decrease in net exports.
- (d) The demand for loanable funds decreases, decreasing the interest rate and increasing the net capital outflow, implying a depreciation of the national currency and a decrease in net exports.

Exercise 8

During the past decade, China's savings were used to finance investment in the United States (U.S.). In other words, the Chinese bought U.S. capital assets.

1. If the Chinese decided that they no longer wanted to buy U.S. assets, what would happen in the US market for loanable funds? What would happen to the US interest rates, the US savings and US investment?
2. What would happen in the market for foreign currency exchange? In particular, what would happen to the value of the USD and to the US trade balance?

Exercise 9

Imagine the case of a country named Pecunia, which in year 2005 experienced a 1 billion Euro deficit in the current account balance. At the same time there were no changes in its official reserves.

1. What was its net capital outflow (NCO)?
2. What was the value of its capital account balance?